

The Scoop 52

Complete Production Launch Plan

Project: Smart Hub Stage → Production

Target traffic: 500,000 public page views / month

Stack: Streamlit · Alpha Vantage · Supabase Pro

Prepared: March 2026

Confidential — Internal Use

1. Executive Summary

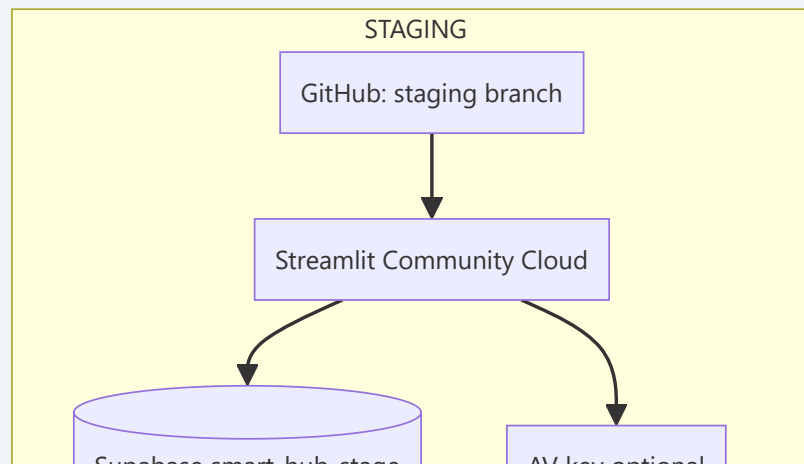
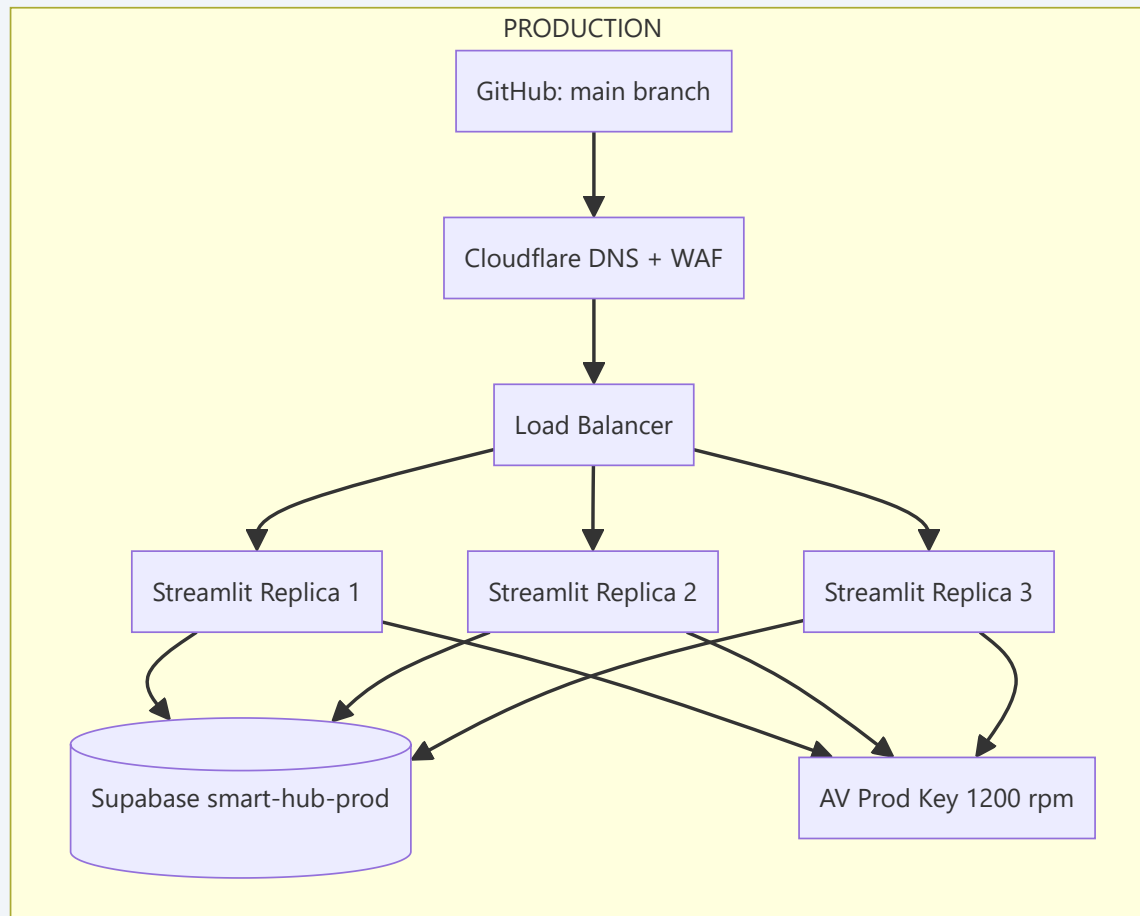
Item	Recommendation
Production host	Render or Railway (Docker, always-on, custom domain)
Staging	Streamlit Community Cloud + smart-hub-stage
Production	2–3 Streamlit replicas + Cloudflare + smart-hub-prod
New monthly spend	~ \$50–75/mo hosting (beyond existing subscriptions)
Already owned	Supabase Pro (~\$25/mo) · Alpha Vantage 1200 rpm (~\$250/mo)
Timeline	2–4 weeks to production-ready
All-in monthly (launch)	~ \$325–350/mo including existing subscriptions

Key insight: With 1200 rpm Alpha Vantage, API rate limits are not the blocker. Focus on always-on hosting, caching headline enrichment, and Supabase prod isolation.

2. Traffic Profile (500k Views / Month)

Metric	Estimate
Daily views	~16,700
Average hourly	~700 views/hr
Peak hour (US market, 3–5× avg)	~2,000–3,500 views/hr
Visits (~2.5 views/visit)	~200,000 visits/month
Avg concurrent Streamlit sessions	~15–25
Peak concurrent sessions	~60–100

3. Environment Layout



Supabase smart-hub-stage

Av Key Optional

4. Environment Comparison

Setting	Staging	Production
Branch	staging	main
Host	Streamlit Community Cloud	Render / Railway / Fly.io
Supabase	smart-hub-stage	smart-hub-prod
Alpha Vantage	Shared or staging key	Dedicated prod key (1200 rpm)
APP_ENV	staging	production
Domain	*.streamlit.app	app.yourdomain.com

5. Complete Pricing Breakdown

5.1 Already Paid (No Incremental Cost)

Service	Plan	Monthly	Notes
Supabase	Pro (org-level)	~\$25	Add smart-hub-prod as second project in same org
Alpha Vantage	1200 rpm premium	~\$250	More than sufficient for 500k views with caching

5.2 New Production Costs — Render (Recommended)

Item	Spec	Monthly
Streamlit × 2	Standard: 2 GB RAM, 1 CPU, always-on	\$50
3rd replica (peak)	Standard instance	+\$25
Cloudflare	Free tier (WAF, DNS, DDoS)	\$0
Domain	.com registrar (if needed)	~\$1
Sentry + UptimeRobot	Free tiers	\$0
Render subtotal (launch)		\$50–75

5.3 Option B: Render Pro (Higher Headroom)

Item	Spec	Monthly
Streamlit × 2	Pro: 4 GB RAM, 2 CPU	\$170

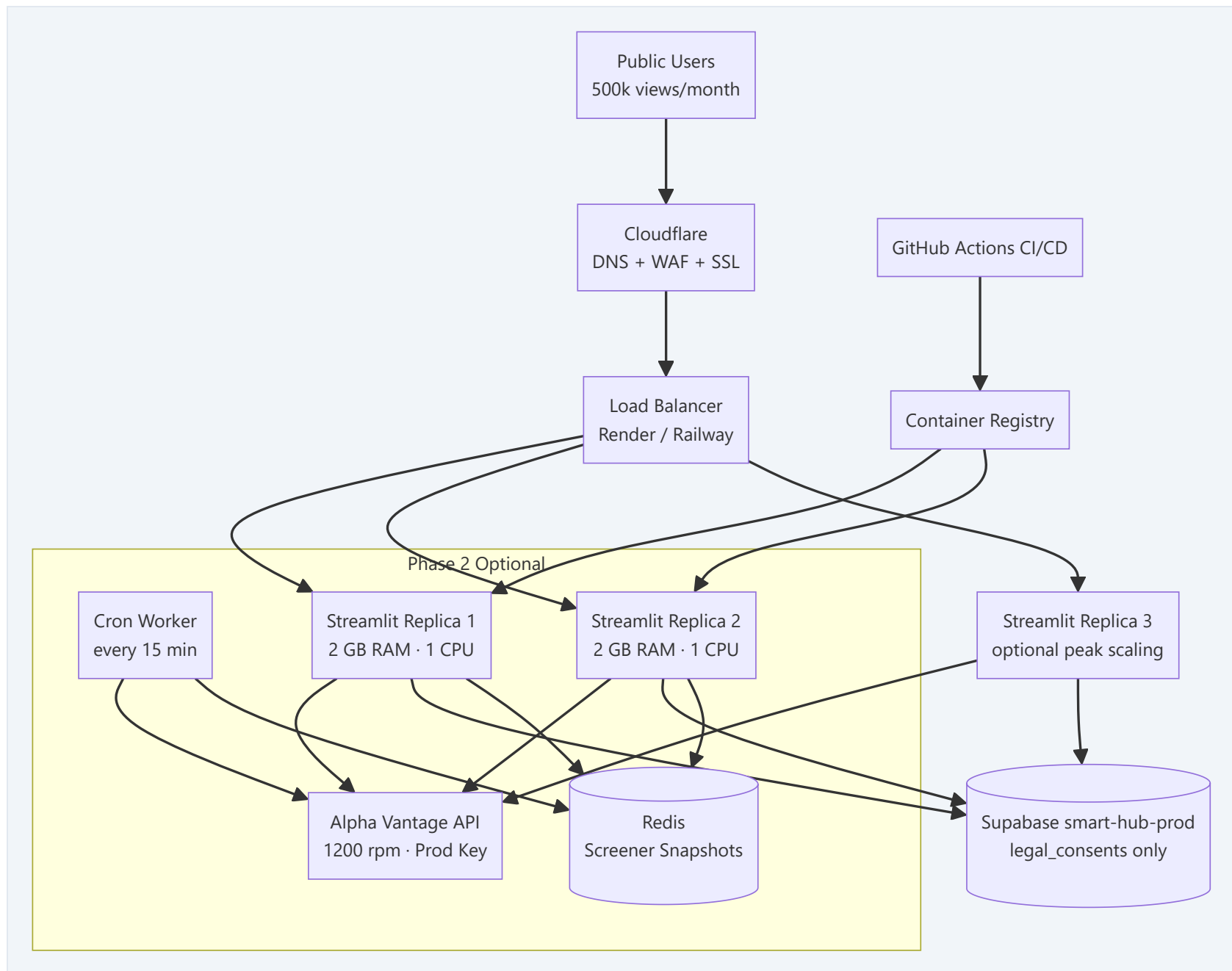
5.4 Phase 2 Add-Ons (Optional v2)

Item	Spec	Monthly
Redis	256 MB–1 GB (Upstash / Render)	\$10–30
Cron worker	15-min precompute job	\$7–25
Cloudflare Pro	Bot management (if abused)	\$20

5.5 Total Monthly Cost

Category	Cost
Supabase Pro (existing)	~\$25
Alpha Vantage 1200 rpm (existing)	~\$250
New prod hosting (launch)	\$50–75
Domain (if needed)	~\$1
Phase 2 Redis + worker (optional)	+\$17–55
All-in at launch	~\$325–350/mo
All-in with Phase 2	~\$345–405/mo

6. Production Architecture (Recommended)



7. Critical Bottlenecks & Fixes

Issue	Impact	Fix	Priority
Headline enrichment not cached	Up to 10 AV news calls per user per screener page	Cache enrich_headline_sentiment (15 min TTL)	Must-have
Per-replica Streamlit cache	Cold-start API spikes on deploy/scale	Rolling deploys; Phase 2 Redis worker	Should-have
Community Cloud for prod	Sleep, 1 GB RAM, no SLA	Docker on Render/Railway	Must-have
Search page abuse	Unbounded per-ticker AV calls	Cloudflare rate limit + Redis cache	Should-have

Alpha Vantage usage at 1200 rpm: Scheduled precompute $\approx$ 110–140 rpm sustained (~10% of capacity). Uncached headline enrichment is the remaining user-scaled risk — fix before launch.

8. Pre-Deploy Checklist

Must-Have Before Launch

1. Dockerfile + .dockerignore
2. Cache headline enrichment (15-min TTL)
3. Create smart-hub-prod Supabase project
4. Prod secrets on host (APP_ENV, DATABASE_URL, ALPHA_VANTAGE_API_KEY)
5. Run migrations: `python admin_tools/run_migrations.py`
6. Cloudflare + custom domain with WebSockets enabled
7. Separate prod Alpha Vantage key from staging

Should-Have (First Month)

1. Search page rate limit (Cloudflare or app-level)
2. Rolling deploys (one replica at a time)
3. Sentry + uptime monitoring
4. Documented `SCREENER_CACHE_VERSION` bump process

9. Week-by-Week Deployment Plan

Week 1 — Foundation

Day	Tasks
1–2	Create smart-hub-prod Supabase project; run migrations; verify consent logging
3–4	Create Dockerfile; test container locally on port 8501
5	Implement headline enrichment caching (critical API fix)
6–7	GitHub Actions CI skeleton (lint → build → deploy trigger)

Week 2 — Production Host

Day	Tasks
8–9	Render: 2× Standard Docker web services; set prod env vars
10	Cloudflare DNS + custom domain; enable WebSockets + Full SSL
11–12	Smoke tests + load test (20 concurrent users); verify all 7 screener pages
13–14	Sentry + UptimeRobot; Render CPU/memory alerts at 80%

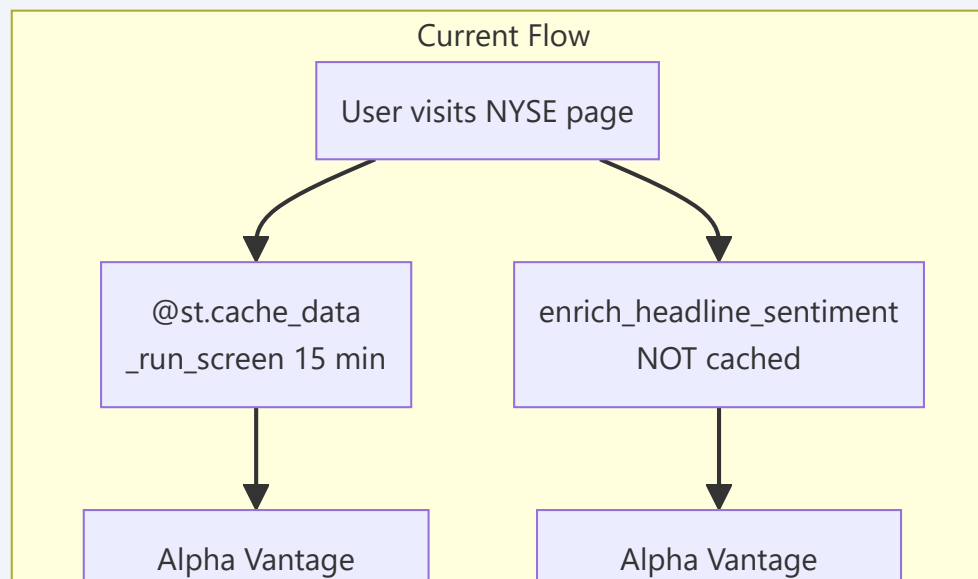
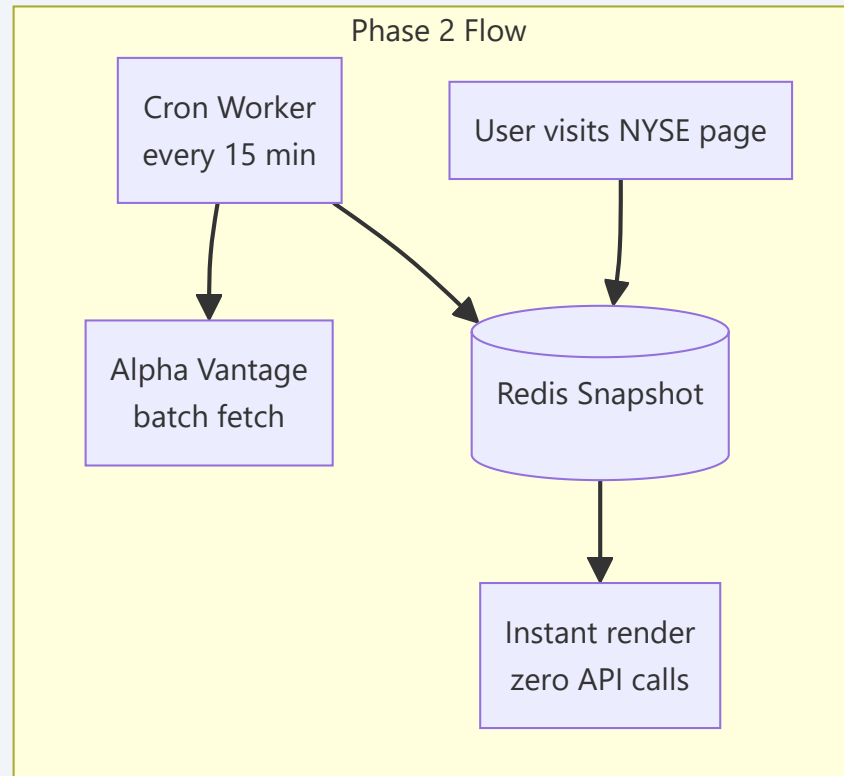
Week 3 — Hardening & Go-Live

Day	Tasks
15–16	Staging/prod isolation audit (DB, keys, APP_ENV, secrets)
17	Search abuse controls (Cloudflare rate limit)
18	Merge staging → main; rolling deploy
19–20	Soft launch; monitor 48 hours
21	Full go-live checklist sign-off

Week 4 — Phase 2 (Optional)

Task	Benefit	Cost
Redis + 15-min precompute worker	Shared snapshots; zero cold-start scans	~\$35–55/mo
Auto-scale 2 → 3 replicas	Handle traffic spikes	+\$25/mo

10. Screener Data Flow (Current vs Phase 2)



price data per symbol

NEWS_SENTIMENT x10

11. Production Secrets Reference

Variable	Value
APP_ENV	production
DATABASE_URL	Supabase prod Session pooler URI + ?sslmode=require
ALPHA_VANTAGE_API_KEY	Prod 1200 rpm key
SCREENER_SYMBOL_LIMIT	1000
CONSENT_DEFAULT_TIMEZONE	America/Chicago

12. Scaling Playbook

Signal	Action	Cost Impact
p95 page load > 15 s	Add 3rd Standard replica	+\$25/mo
CPU > 70% sustained	Upgrade to Pro replicas (4 GB)	+\$120/mo
Inconsistent screener timestamps	Add Redis worker	+\$35–55/mo
Search abuse / AV spikes	Cloudflare bot rules + search cache	\$0–20/mo

13. Risk Register

Risk	Likelihood	Mitigation
Uncached headline news calls spike AV	High (without fix)	Cache enrichment — Week 1
Community Cloud used for prod by mistake	Medium	Separate domain + APP_ENV banner
Deploy cold-starts all replicas	Medium	Rolling deploy; Phase 2 Redis
WebSocket breaks behind proxy	Medium	Cloudflare WebSockets on; Full SSL
Staging data in prod DB	Low	Separate Supabase project + audit
Public search abuse	Medium	Cloudflare rate limits

14. Launch Day Runbook

T-24 Hours (Pre-Launch)

- Prod Supabase migrated; test consent row

- Dockerfile builds locally
- Headline enrichment cached
- 2 Render replicas healthy
- Cloudflare DNS proxied; WebSockets OK
- Sentry + UptimeRobot active
- Staging branch $\neq$ main secrets

T-0 (Launch)

- Merge staging $\rightarrow$ main
- Watch Render deploy (rolling, one instance at a time)
- Verify <https://app.yourdomain.com>
- Accept Terms $\rightarrow$ confirm legal_consent row in prod
- Spot-check NYSE, NASDAQ, Crypto pages

T+48 Hours (Post-Launch)

- Sentry: zero recurring errors
- AV: no sustained rate-limit responses
- Render CPU < 60% avg
- Page load p95 acceptable
- No consent logging failures in Supabase

15. Platform Comparison

Platform	Pros	Cons	Launch Cost
Render ✓	Docker-native, simple secrets, health checks	Avoid free tier (cold starts)	~\$50/mo
Railway	Fast setup, good DX	Usage-based; less predictable	~\$40–80/mo
Fly.io	Global edge, fine-grained scaling	More ops (fly.toml)	~\$30–60/mo

Recommendation: Render, 2 $\times$ Standard Docker services — best balance of simplicity and predictable cost for a public Streamlit app at 500k views/month.

16. Bottom Line

Supabase Pro and Alpha Vantage 1200 rpm are already in place. Production deployment adds **~\$50–75/mo in hosting** and a **2–3 week execution plan**:

1. **Week 1:** Prod Supabase, Dockerfile, headline cache fix
2. **Week 2:** Render (2 replicas), Cloudflare, domain, monitoring
3. **Week 3:** Hardening, load test, go-live
4. **Week 4 (optional):** Redis worker for replica consistency

Total all-in at launch: ~\$325–350/month including existing subscriptions.